



Metabolic Syndrome

Metabolic syndrome involves having at least 3 out of 5 health conditions that increase your risk of cardiovascular disease, stroke and Type 2 diabetes. It can cause other complications as well. Each condition is treatable with lifestyle changes and/or medication.

Overview

What is metabolic syndrome?

Metabolic syndrome is a group of conditions that together increase your risk of **cardiovascular disease**, **Type 2 diabetes** and **stroke**. It can lead to other health problems as well, like conditions related to plaque buildup in artery walls (**atherosclerosis**) and organ damage.

Other names for metabolic syndrome include:

- Syndrome X.
- Insulin resistance syndrome.
- Dysmetabolic syndrome.

Criteria for metabolic syndrome

A person meets the criteria for metabolic syndrome if they have at least three of the following:

- **Excess abdominal weight:** A waist circumference of more than 40 inches in **males** and 35 inches in females
- **Hypertriglyceridemia:** Triglyceride levels that are 150 milligrams per deciliter of blood (mg/dL) or greater.
- **Low levels of HDL cholesterol:** HDL cholesterol of less than 40 mg/dL in males or less than 50 mg/dL in females.
- **Elevated blood sugar levels:** Fasting blood sugar level of 100 mg/dL or greater. If it's 100 to 125 mg/dL, you have **prediabetes**. If it's over 125 mg/dL, you likely have Type 2 diabetes.
- **High blood pressure:** Blood pressure values of systolic 130 mmHg or higher (the top number) and/or diastolic 85 mmHg or higher (the bottom number).

All of these conditions individually increase your risk of cardiovascular disease, Type 2 diabetes and stroke. But when you have three or more, your risk increases significantly. You should see a diagnosis of metabolic syndrome as a warning sign to try to change aspects of your health to lower your risk.

How common is metabolic syndrome?

Metabolic syndrome is common in the United States. About 1 out of every 3 adults have it.

Symptoms and Causes

What are the symptoms of metabolic syndrome?

Not all aspects of metabolic syndrome cause symptoms. So, your symptoms will vary based on which of the five conditions you have. For example, high blood pressure, high triglycerides and low HDL cholesterol usually don't cause symptoms.

High blood sugar (hyperglycemia) can cause symptoms for some people, like:

- Darkened skin in your armpits or the back and sides of your neck (acanthosis nigricans).
- Blurred vision.
- Increased thirst (polydipsia).
- Increased urination, especially at night.
- Fatigue.

See a healthcare provider if you have these symptoms.

What causes metabolic syndrome?

Several factors contribute to the development of metabolic syndrome — and it's a complex web of factors. But researchers think **insulin resistance** is the main driver behind the syndrome.

Insulin resistance happens when cells in your **muscles**, fat and **liver** don't respond as they should to **insulin**, a hormone your **pancreas** makes that's essential for life and regulating **blood glucose (sugar)** levels.

For several reasons, your muscle, fat and liver cells can respond inappropriately to insulin. This means they can't efficiently take up glucose from your blood or store it. This is insulin resistance. As a result, your pancreas makes more insulin to try to overcome your increasing blood glucose levels. This is called **hyperinsulinemia**.

If your body can't produce enough insulin to effectively manage your blood sugar, it leads to high blood sugar (hyperglycemia) and prediabetes or Type 2 diabetes. Insulin resistance and hyperinsulinemia can also contribute to:

- Obesity.
- Cardiovascular disease.
- Fatty liver disease.
- Polyendocrine metabolic ovarian syndrome (PMOS).

The following can all contribute to insulin resistance:

- **Excess weight around your abdomen or having obesity:** Body fat releases chemicals (called proinflammatory **cytokines**) that dampen the effect of insulin. The more excess body fat you have, the more it can negatively affect how insulin works. Studies show that excess body fat around your abdomen, in particular, increases your risk of insulin resistance. Excess **visceral fat** (fat around your organs) causes more insulin resistance than excess **subcutaneous fat** (fat under your skin). But they both play a role in metabolic syndrome.
- **Lack of physical activity:** Your muscles use a lot of glucose and stored glucose (**glycogen**) to function. Physical activity makes your body more sensitive to insulin and builds muscle that can absorb more blood glucose. A lack of physical activity can have opposite effects and cause insulin resistance.
- **Certain medications:** Certain medications can cause insulin resistance, including corticosteroids, some blood pressure medications, certain **HIV** treatments and some psychiatric medications.

- **Genetics:** The **genes** you inherited from your biological parents can contribute to insulin resistance. They can also contribute to having obesity, high blood pressure and high cholesterol.

Diagnosis and Tests

Image content: This image is available to view online.

View image online (<https://my.clevelandclinic.org/-/scassets/images/org/health/articles/10783-metabolic-syndrome>)

Metabolic syndrome increases your risk of cardiovascular disease, Type 2 diabetes and stroke. You have metabolic syndrome if you meet at least three of the criteria.

How is metabolic syndrome diagnosed?

A healthcare provider will do a **physical exam** and order **blood tests** if they think you might be at risk for or have metabolic syndrome. They'll check your blood pressure and may measure the circumference around your waist.

They'll order blood tests, like:

- **Lipid panel:** This panel includes four different cholesterol measurements and a measurement of your triglycerides.
- **Basic metabolic panel (BMP):** This panel measures eight substances in your blood and gives an overall view of your health.
- **Fasting glucose test:** A BMP includes a blood glucose reading, but if you didn't fast for the BMP, your provider may have you get a blood test that checks your blood sugar after fasting for eight to 12 hours.

If you have at least three of the five criteria based on the results of these tests and the exam, you'll have metabolic syndrome.

These blood tests are typically routine tests. So, your provider may tell you that you have metabolic syndrome (or are at risk for certain health conditions) after routine tests.

Management and Treatment

How is metabolic syndrome treated?

The main goals of treating metabolic syndrome are to lower your risk of heart disease and Type 2 diabetes if you don't already have them. Treatment can involve medications and/or lifestyle changes.

Lifestyle changes to manage metabolic syndrome

Lifestyle changes are key to managing the conditions that contribute to metabolic syndrome. Changes include:

- **Maintaining or working toward a weight that's healthy for you:** Your healthcare provider may recommend trying to lose excess weight. One study revealed that losing 7% of excess weight can reduce the onset of Type 2 diabetes by 58%.
- **Getting regular exercise:** Physical activity has numerous benefits. It helps combat insulin resistance, can help keep your cardiovascular system healthy and may help you lose weight if needed. Any

increase in physical activity is helpful. But before starting an exercise program, ask your provider about what level of physical activity is right for you.

- **Eating heart-healthy foods:** Your provider or nutritionist may recommend avoiding eating excessive amounts of carbohydrates (which stimulate excess insulin production) and eating less unhealthy fat, sugar, red meats and processed starches. Instead, they'll likely recommend eating a diet of whole foods that includes more vegetables, fruits, whole grains, fish and lean poultry. The **Mediterranean diet** is one example of a heart-healthy diet.
- **Getting quality sleep:** Quality sleep is vital to overall health. A lack of sleep and **sleeping disorders** (like **sleep apnea**) can worsen metabolic syndrome or contribute to its development. If you're having problems sleeping, talk to your healthcare provider. They can do tests and suggest treatments or changes to your sleeping routine.
- **Avoiding or quitting smoking:** Smoking can decrease your HDL cholesterol and increase blood pressure. It also damages your blood vessels, which can lead to coronary artery disease. If you smoke, try to quit.
- **Managing stress:** High levels of cortisol (the "stress hormone") over long periods of time can increase triglycerides, blood sugar and blood pressure. Find strategies to manage your stress, like exercise, yoga, mindfulness or breathing exercises.

Medications and treatments for managing metabolic syndrome

Various medications and treatments can help manage the conditions that contribute to metabolic syndrome. They include:

- **Cholesterol medications:** Statins (HMG CoA reductase inhibitors) are prescription medicines that people take to bring their cholesterol down to healthy levels.
- **Blood pressure medications:** These medications (antihypertensives) are prescription medicines that bring your blood pressure down in various ways. Examples include thiazide, ACE inhibitors and calcium channel blockers.
- **Oral diabetes medications:** These medications work in various ways to lower your blood sugar. The most common medication is **metformin**, a **biguanide**.
- **Bariatric surgery:** Bariatric surgery (weight loss surgery) is a category of surgical operations intended to help people with obesity lose weight. Your provider may recommend bariatric surgery if other weight loss methods haven't worked and if obesity poses a greater risk to your health than surgery.
- **Sleeping disorder treatments:** If you have a sleeping disorder, certain treatments can help, like a CPAP machine for sleep apnea or **sleeping pills** for insomnia.
- **Psychotherapy:** "Psychotherapy" (talk therapy) is a term for a variety of treatment techniques that aim to help a person identify and change unhealthy emotions, thoughts and behaviors. Psychotherapy may help you manage stress or understand and change unhealthy behaviors related to eating, for example.

Can you reverse metabolic syndrome?

Yes, it's possible to reverse metabolic syndrome. Lifestyle changes can do a lot to improve your health. Medications can help as well. Your healthcare provider will work with you to find the best plan for you.

Outlook / Prognosis

What are the possible complications of metabolic syndrome?

Metabolic syndrome can lead to a wide range of complications, including:

- Heart disease.
- Aortic stenosis (when the aortic valve in your heart narrows and blood can't flow normally).
- Atrial fibrillation (Afib).
- Thromboembolic disease, like venous thromboembolism.
- Stroke.
- Organ damage, especially damage to your pancreas, liver, gallbladder and kidneys.
- Certain cancers, like colon cancer, breast cancer and prostate cancer.
- Type 2 diabetes.
- Long-term inflammation and problems with your immune system.
- Erectile dysfunction.
- Pregnancy complications, such as preeclampsia, eclampsia and gestational diabetes.
- Issues with thinking and memory.

The good news is that it's possible to reverse metabolic syndrome with lifestyle changes and medications. The sooner you can make changes to protect your health, the better.

Prevention

Can I prevent metabolic syndrome?

You can't change all the factors that contribute to metabolic syndrome, like your genetics and age. But the lifestyle changes that can help treat metabolic syndrome are the same steps that can help prevent it.

If you have a family history of diabetes, high blood pressure or high cholesterol, be sure to tell your healthcare provider.

It's also important to schedule routine provider visits. Your provider can keep track of your cholesterol, triglyceride, blood pressure and blood sugar levels. The sooner they can catch any issues, the sooner they can recommend lifestyle changes and treatments to reduce your risk.

Living With

When should I see my healthcare provider?

If you have metabolic syndrome, it's important to get ongoing care. You should see your healthcare provider for the following:

- **To monitor the condition:** You may need to measure your blood pressure regularly or have routine blood tests to monitor your triglyceride and cholesterol levels. Be sure to keep all of your healthcare appointments.
- **For questions about your treatment plan:** Tell your provider if you have side effects from your medications or if you want to stop taking them. Also, ask any questions about lifestyle changes, like diet or exercise plans.

A note from Cleveland Clinic

You may be overwhelmed to learn you have metabolic syndrome. Know that there are several strategies to manage or reverse it. Your healthcare provider will be with you every step of the way. Lean on them for support and advice. They're available to help you.

Experts You Can Trust

✔ **Medically Reviewed.** Last updated on 09/13/2023.

References

Cleveland Clinic's health articles are based on evidence-backed information and review by medical professionals to ensure accuracy, reliability and up-to-date clinical standards.

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